Missing Women in Research

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May 6, 2026

ABSTRACT.

We provide the first comprehensive, country-wide analysis of gender gaps in academic trajectories after PhD graduation, covering 30 years and all academic disciplines. Combining data on all French PhD graduates and all academic publications, we document persistent and sizable negative gender gaps. Women are less likely to ever publish than men and publishing women publish less publications than publishing men in all fields of research and career stages. These negative gaps have not declined over time, despite improvements in female representation. They are worse in STEM at the extensive margin, but not at the intensive margin. Our findings provide evidence of substantial and persistent gendered barriers operating early in academic career. We estimate that removing these barriers would increase the population of women in research by about 27%.

Keywords: Gender Inequality, Career Trajectory, Graduate Education

JEL: I23, J16, J24

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1 Introduction

Despite improvements in the past 40 years, women are still severely underrepresented in academia, as documented in the *UNESCO Science Report*. In France, the proportion of female university faculty has not reached 30% yet. This underrepresentation persists despite the fact that women are now, on average, more educated than men: in 2017, 55 % of French graduates in higher education were women. Is just a question of delay, then? Will the proportion of female faculty track, with a lag, the proportion of women among higher education graduates? Or are there significant barriers preventing, somehow, highly educated women from becoming successful academics? How does the answer to these questions vary across academic disciplines and time? Even though the literature has made progress on these key questions, we still lack comprehensive answers.

In our study, we focus on a critical stage: the transition from a PhD to an academic career. We provide the first comprehensive, country-wide analysis of gender differentials in academic trajectories following PhD graduation, covering 30 years and all academic disciplines. To do so, we merge data on all PhD graduations in France between 1988 and 2021 from *theses.fr* with data on all academic publications until 2022 from *Scopus*. Our final sample has information on the academic trajectories of 250,889 PhD graduates, 40 % of which are women. This is, to our knowledge, the first study of academic gender gaps at this scale.

We develop our analysis in several stages. We first document how the proportion of women among PhD graduates has evolved over time and across disciplines. Encouragingly, female representation increased significantly in all disciplines. Women remain severely underrepresented, however, in *STEM (Sciences, Technology, Engineering, and Mathematics)*. By contrast, they are now slightly overrepresented in *Humanities and Law*, *Biological and Earth Sciences*, and *Social Sciences*. We then analyze academic trajectories post-graduation. Does a higher female representation among PhD graduates translate into smaller gender gaps? Once gender parity is reached at the PhD level in a discipline, do male and female graduates tend to follow similar academic trajectories?

To answer this question we collected data on the name, discipline, year of defense, university affiliation and academic publications of each PhD graduate, as well as of their PhD supervisor(s).[1] We infer gender from first names. Overall, 54% of PhD graduates have no academic publication.

[1] For each publication, we have information on year of publication, publication type and outlet, and names and affiliations of authors.

This motivates separate analyses of the extensive and intensive margins of research. We thus look at both the likelihood of publishing at least once and the number and impact of publications conditional on publishing. We also distinguish two career stages: early career, until 4 years post-graduation, and mid-career, between 5 and 10 years post-graduation. We perform regressions for the full sample and separately for subsamples of the four main fields of research. We analyze both raw gender gaps, capturing differences in means between women and men, and adjusted gender gaps, estimated from regressions that control for university-discipline-year fixed effects and supervision characteristics. Finally, we study how gender gaps evolve over time.

Overall, we document persistent and sizable negative gender gaps in academic trajectories. Women are less likely to ever publish than men and publishing women publish less publications than publishing men in all fields of research and at both career stages. These negative gaps have not declined over time. They tend to be worse in STEM at the extensive margin, on the likelihood to ever publish, but not at the intensive margin, on the number of publications conditional on publishing. Our findings thus provide evidence of substantial and persistent gendered barriers operating early in academic career.

By contrast, we detect no difference in the impact of publications of publishing men and publishing women in all fields except Biological and Earth Sciences, in which the gender gap in impact is negative but quantitatively small. In addition, raw gaps are generally less negative, a sign of positive sorting of women across space, time and discipline.

Finally, we use our regressions to estimate the number of missing women in academia, building on studies of missing women in poor countries, see Sen (1990), Coale (1991), Anderson and Ray (2010). We compute the number of women who would become academically active if the barriers affecting women in the post-PhD transition were removed. Based on the field-specific regressions of the mid-career extensive margin, we estimate that 5,737 women are missing in academia in France across all disciplines in the 25 years period from 1988 to 2012. This represents 27% of the population of publishing female PhD graduates. Note that these numbers are computed by conditioning on the existing population of PhD graduates. They would be larger if we also accounted for gender imbalances at the PhD level.

Our analysis contributes to the literature on gender and academia. Earlier studies have documented the importance and persistence of the underrepresentation of women in academia, see Ginther and Kahn (2004), Ceci (2011), Ceci (2014), Meyer et al. (2015), Huang (2020). Many studies focus on STEM, where this underrepresentation is particularly severe. Other studies have documented the fact that men appear to be more productive than women in research, and notably in economics (Ginther and Kahn, 2004; Barbezat, 2006; McDowell et al., 2006; Ductor et al., 2023; Conley et al., 2016), science (Patsali et al., 2024; Stephan and Levin, 1997), and medicine (Rachid et al., 2021).

We also contribute to the literature on career paths in academic jobs. Recent works have identified several barriers to carrier promotion of women in economics linked to recognition of group work, citations, fellowships, and writing criteria (Sarsons, 2017; Eberhardt et al., 2023; Card et al., 2019, 2022; Hengel, 2022). It has also been shown that women are substantially underrepresented in senior positions in economics (Bosquet et al., 2019). We contribute to this literature by analyzing publication gaps at different carrier stages and across disciplines.

A few studies combine data on PhD graduations and publications, as we do. Existing studies, however, focus on specific disciplines and shorter time scales, and do not systematically study gender gaps. Gaule and Piacentini (2018) analyze the academic trajectories of PhD graduates in chemistry in the US who defended between 1999 and 2008. They find a strong impact of the gender match between a PhD graduate and their supervisor. Conley and Önder (2014) study the research productivity of PhD graduates in economics in the US and Canada who defended between 1986 and 2000. They find that the academic rank of the department from which a PhD student graduates provides a surprisingly poor predictor of the student's research success. Corsini et al. (2022) analyze the determinants of PhD graduates' scientific productivity, based on the population of PhD graduates in France in STEM over the period 2000 - 2014. Patsali et al. (2024) analyze how research independence from the PhD supervisor(s) affects academic outcomes for a similar population of PhD graduates in France in STEM over the period 1995 - 2013.

Our main contributions with respect to these studies are the focus and scale of our analysis. We provide the first analysis of gender gaps covering 30 years and all academic disciplines. We look at three dimensions of academic productivity - probability to ever publish, number and impact of publications - at both early and mid-career. We provide a detailed anatomy of the gender gaps and of how they vary across fields and time. This notably allows us to understand how STEM - and economics - compare with other less studied disciplines.

The remainder of the paper is organized as follows. We describe the data in Section 2. We provide key descriptive statistics in Section 3. We present our empirical strategy and our main findings in Section 4. We compute numbers of missing women in Section 5. We conclude in

Section 6.

2 Data

We combine data from two sources: *theses.fr* on PhD graduations and *Scopus* on academic publications. We next describe the data and our sampling restrictions: the detailed cleaning and variable construction process for Phd graduations and publications are described in Appendix A and Appendix B respectively.

2.1 PhDs

We retrieved data from *theses.fr* on all PhD theses defended in French universities from 1988 to 2021. For each PhD thesis, we have information on discipline of study, defense year, university affiliation, and first and last name of the PhD graduate and the supervisor(s). *theses.fr* is a centralized, public platform that collects data on all PhD graduations in the French academic system. French universities have the legal obligation to report information on PhD graduations. While some spelling errors and reporting delays are unavoidable, this database is considered as being, overall, comprehensive and reliable.[2] The initial sample contains 407,260 observations.

We focus on PhD theses with one or two supervisors and with non-missing information on discipline and names, reducing the sample to 393,259 observations.[3]

We distinguish 22 academic disciplines, following the Australian and New Zealand Standard Research Classification. We exclude theses defended in *Health and Medical Sciences*, because of a misclassification problem specific to this discipline.[4] This further reduces the sample to 335,796 observations. We aggregate the 21 remaining disciplines in 4 broad fields of research: *Humanities and Law*; *Biological and Earth Sciences*; *Science, Technology, Engineering, and Mathematics (STEM)* and *Social Sciences*. In our empirical analysis below, we estimate regressions on the overall sample as well as separate regressions per field. We also pay special attention to *Economics*.

One issue with university affiliations is that France has seen some significant institutional evolution in the past twenty years, including a number of university mergers. We keep track of

[2]We chose 2021 as the last year to minimize problems of reporting delays. See https://theses.fr/ for more information on data collection and on the underlying institutional arrangement.

[3]Theses with three supervisors or more represent less than 2% of the original sample.

[4]To become Medical Doctors, French medical students must defend a practical thesis ("Thèse d'exercice") at the end of their studies. These practical theses have very different requirements from PhD theses. However, theses.fr did not distinguish between the two kind of theses until the 2000s.

these evolutions and notably distinguish between universities before and after mergers.

2.2 Publications

We next assemble information on academic publications of PhD graduates and PhD supervisors. To do so, we extract bibliometric data from *Scopus* on all publications until 2022 for which the first name and last name of one author are identical to the first name and last name of a PhD graduate or a PhD supervisor. Since there is a high risk of mismatch when the first and last names are very common (such as "Sarah Lopez" or "Philippe Morin"), we remove observations with overly common names of PhD graduates and supervisors, reducing the sample to 304,367 PhD theses. For each academic publication we have information on publication type, journal, year of publication, authors and their affiliations.

2.3 Gender

We identify the gender of PhD graduates and PhD supervisors using well-established methods based on first names. In a first step, we use data from the French statistical institute, *INSEE*, on the numbers of boys and girls born in France between 1940 and 2020 with a given first name. We associate a gender with a first name when at least 95 % of individuals with this name share the same gender. In a second step, and to broaden coverage to non-French PhD graduates and supervisors, we use similar data from Australia, Canada, Spain, Sweden, the UK, and the US. In the end, we identify the gender of 93 % of the PhD graduates and 95 % of the supervisors. We consider PhD graduates with an identified gender and with at least one supervisor with an identified gender, further reducing the number of observation to a final sample of 274,944 PhD theses over the period 1988-2021.

Table C1 in the Online Appendix describes the number of PhD graduates per discipline. The 274,944 Phd students are distributed as follows: 20 % in Humanities and Law, 17 % in Biological and Earth Sciences, 44 % in STEM and 18 % in Social Sciences. Among PhD graduates in Social Sciences, 19 % have graduated in Economics.

2.4 Early and mid-career

In our analysis of academic trajectories, we distinguish two periods: early career and mid-career. We define early career as every year until the fourth year after graduation. E.g., for a researcher graduating in 2000, publications in every year up to 2004 are counted in early career. We define mid-career as the period between five and ten years after graduation, both years included. E.g., for a researcher graduating in 2000, publications in every year between 2005 and 2010 are counted in mid-career. To study academic trajectories post-graduation, we need to trim the last years of our sample - recall that we observe academic publications until 2022. The final sample for the early career analysis is thus composed of all PhD graduates who graduated between 1988 and 2018, corresponding to 250,889 observations. The final mid-career sample is composed of all PhD graduates who graduated between 1988 and 2012, corresponding to 192,441 observations.

2.5 Research outcomes

We analyze both the extensive and the intensive margin of academic research. We consider three outcomes: *Any Publi*, *Publications* and *Impact*. *Any Publi* is a binary variable, equal to one if the PhD graduate has at least one academic publication referenced in *Scopus* over the period of interest and to zero otherwise. For all PhD graduates for whom *Any Publi*=1, we then compute two measures of research productivity. *Publications* is equal to the fractional publication count, that adds up, over all publications, one over the number of authors. Dividing by the number of authors is a standard way to allocate credit for shared authorship. Our results are robust to using the undivided publication count instead, see Section C.1 of the Online Appendix. Our third outcome, *Impact*, measures the average impact of the PhD graduate's publications. We build this measure from the Article Influence Score (AIS) of the publication outlet, a classical measure of impact factor (see Bagues et al., 2017). *Impact* is then equal to the AIS-weighted fractional publication count divided by the fractional publication count.

Formally, consider PhD graduate i with set of publications P_i. Given a publication $p \in P_i$, let n_p denote the number of authors of the publication and AIS_p the Article Influence Score of the publication outlet. Then, we formally define for each Phd graduate i

$$\begin{aligned} Any\ Publi &= 1 \text{ if } P_i \neq \emptyset \text{ and } 0 \text{ if } P_i = \emptyset \\ Publications &= \sum_{p \in P_i} \frac{1}{n_p} \\ Impact &= \frac{\sum_{p \in P_i} \frac{AIS_p}{n_p}}{\sum_{p \in P_i} \frac{1}{n_p}} \end{aligned}$$

Finally, in all regressions we standardize *Publications* and *Impact* to have mean 0 and standard deviation 1 within discipline and career stage. More precisely, we compute the mean

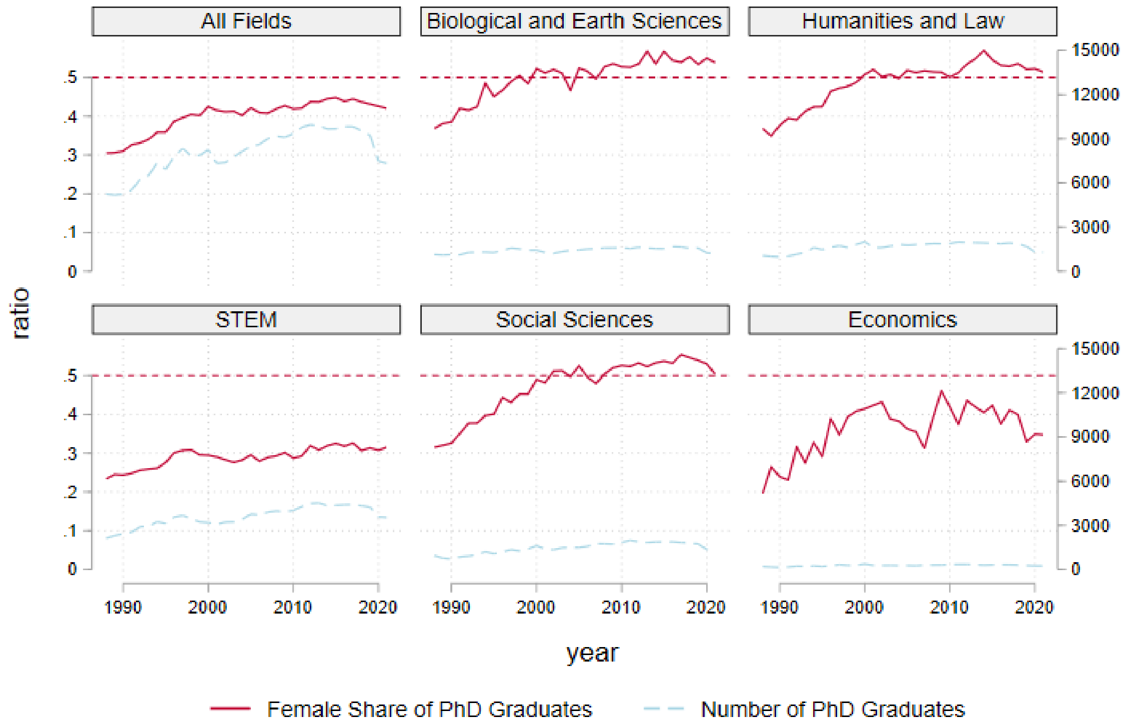

Figure 1: Share of female PhD graduates

and standard deviation of *Publications* and *Impact* among all publishing PhD graduates for each of the 21 disciplines and 2 career stages. We then standardize each outcome by subtracting the mean and dividing by the standard deviation. One advantage of this normalization is that it makes the gender gap estimates directly comparable across fields.

3 Descriptive statistics

We now provide some key descriptive statistics of the data. We first describe how the proportion of female PhD graduates varies across time and field. We then discuss systematic differences in research productivity across fields.

3.1 Female representation among PhD graduates

Figure 1 depicts how the proportion of female PhD graduates (in solid red) and the total number of PhD graduates (in dotted blue) vary over time in the full sample, Humanities and Law, Biological and Earth Sciences, STEM, Social Sciences and Economics. The red dotted line corresponds to a 50% proportion of female PhD graduates.

The evolution of female representation displays a similar pattern in Humanities and Law, Biological and Earth Sciences, and Social Sciences. Women were underrepresented in the 1990s in these three fields, with a share lying between 30 and 40%. Female representation then improved, and these three fields reached gender parity in the 2000s. Women then became slightly overrepresented among PhD graduates in the three fields in the 2010s. By contrast, women are still severely underrepresented among STEM PhD graduates, with the female share in the best year equal to 33% in 2017. Female representation did increase over time in STEM, although at a lower rate than in the other fields. Overall, Figure 1 reveals important differences in female representation between STEM and the other academic fields.

Figure C1 in the Online Appendix depicts how the share of female PhD graduates and the number of PhD graduates vary over time in each of the 21 academic disciplines. Remarkably, female representation increased over time in all disciplines, with the possible exception of Mathematics. Within Social Sciences, Economics is the only discipline that never reached gender parity. Following good progress in the 1990s, from a low female share of 20% in 1988, female representation among PhD graduates in Economics in France appears to fluctuate below a 45% ceiling.

3.2 Differences in research productivity across fields

Table C2 in the Online Appendix describes how various measures of research productivity vary across fields and career stages. We report the proportion of publishing PhD graduates for each subsample and the average of the number of undivided publications, of the number of coauthors, and of our two main measures, *Publications* and *Impact*, among publishing PhD graduates.

We see that fields have very different baseline levels of publications, coauthorship and impact, reflecting different academic practices and norms. The likelihood to ever publish, the numbers of publications and of coauthors at both career stages are globally consistent with the following ordering: Humanities and Law $<$ Social Sciences $<$ Biological and Earth Sciences $<$ STEM. The proportions of PhD graduates with at least one publication four years after graduation are, respectively, $15\% < 25\% < 50\% < 66\%$. And a publishing PhD graduate at early career has on average $0.9 < 1.6 < 9.2 < 11.4$ coauthors. The number of undivided publications are, respectively, equal to 2.3, 3.0, 8.7 and 8.4, with Biological & Earth Sciences marginally exceeding STEM. Interestingly, differences among fields are much lower, and the previous ordering is not preserved, when looking at fractional publications counts. Allocating credit equally among coauthors, the respective average numbers of *Publications* at early career are 1.73, 1.71, 1.40

and 1.82.

In terms of impact, the two indices at early and mid-career are consistent with a slightly different ordering, Humanities and Law < Social Sciences < STEM < Biological and Earth Sciences. For instance, the average *Impact* at early career for publishing PhD graduates is equal to 0.17 in Humanities and Law, 0.36 in Social Sciences, 0.59 in STEM and 1.63 in Biological and Earth Sciences. Researchers tend to publish in more influential outlets in Biological and Earth Sciences than in STEM.

Economics appears to be fairly representative of other Social Sciences in terms of probability to publish and numbers of publications and coauthors conditional on publishing. Publications in Economics have more impact, however, than in other Social Sciences. Average *Impact* at early career for publishing PhD graduates in Economics is equal to 0.68.

The presence of systematic differences in research productivity across fields and disciplines motivates the normalization of *Publications* and *Impact* by discipline and career stage in our regressions below.

4 Empirical analysis

We now analyze differences in academic trajectories between female and male PhD graduates. We look at the three research outcomes, *Any Publi* and, if *Any Publi*=1, *Publications* and *Impact*, at both career stages, overall, for each of the four fields of research, and for Economics. We first compute raw estimates, regressing outcome on gender of the PhD graduate only. These provide differences in average outcomes between female and male PhD graduates. In our main specification, we then include university-discipline-year fixed effects and controls related to supervision.

4.1 Raw gender gaps

Table 1 presents the estimates of gender differences in academic outcomes in the overall sample (first column), in the four fields (second to fifth column), and in Economics (last column).

On the overall sample, we see that women have a lower probability to ever publish and publishing women publish less papers than publishing men. These negative gender gaps are present at early and mid-career and are quantitatively significant. The early career gender gap at the extensive margin is -9 p.p., from an average probability to ever publish for men equal to

Table 1: Raw gender gaps

	All Fields	Humanities and Law	Biological and Earth Sc.	STEM	Social Sciences	Economics
			Before t+4			
% Female PhD graduates	40%	49%	50%	29%	48%	37%
Any Publi	**-0.0919***** (0.00769)	**-0.0144***** (0.00463)	**0.0559***** (0.0134)	**-0.0724***** (0.00571)	**0.0102*** (0.00527)	**-0.0180*** (0.00965)
Av. Male Any Publi	0.488	0.153	0.476	0.678	0.241	0.277
Relative Gender Gap	-19%	-9%	12%	-11%	4%	-6%
# PhD graduates	250889	51692	44289	109867	45041	8504
if Any Publi = 1						
Publications	**-0.180***** (0.0162)	**-0.237***** (0.0514)	**-0.358***** (0.0231)	**-0.241***** (0.00910)	**-0.238***** (0.0298)	**-0.306***** (0.0602)
Impact	**0.0101** (0.00929)	**-0.00365** (0.0227)	**-0.0267**** (0.0133)	**0.0350***** (0.0112)	**-0.0389**** (0.0196)	**-0.0313** (0.0394)
# PhD graduates *if Any Publi = 1*	113067	7522	22310	72174	11061	2301
			Between t+5 and t+10			
% Female PhD Graduates	39%	48%	48%	28%	47%	37%
Any Publi	**-0.0909***** (0.00576)	**-0.0194***** (0.00522)	**0.00685** (0.0151)	**-0.117***** (0.00426)	**-0.00700** (0.00600)	**-0.0271**** (0.0117)
Av. Male Any Publi	0.376	0.155	0.396	0.492	0.230	0.262
Relative Gender Gap	-24%	-13%	2%	-24%	-3%	-10%
# PhD graduates	192441	40233	34718	83548	33942	6625
if Any Publi = 1						
Publications	**-0.258***** (0.0223)	**-0.158***** (0.0420)	**-0.450***** (0.0228)	**-0.329***** (0.0131)	**-0.318***** (0.0357)	**-0.381***** (0.0549)
Impact	**0.0200*** (0.0105)	**0.00416** (0.0240)	**-0.0159** (0.0155)	**0.0496***** (0.0136)	**-0.00886** (0.0306)	**-0.00369** (0.0885)
# PhD graduates *if Any Publi = 1*	65546	5845	13639	38361	7701	1667

Notes: The table reports raw gender gaps in publication outcomes for PhD graduates, estimated separately for two time windows: before t+4 and between t+5 and t+10, where t denotes the year of PhD graduation. Any Publi is a binary variable equal to 1 if the PhD graduate has at least one publication in the given period. Publications and Impact are continuous outcomes measured conditional on having at least one publication. All coefficients report the female dummy coefficient from OLS regressions, capturing the raw gender gap relative to male PhD graduates. Relative Gender Gap is computed as the ratio of the female coefficient to the average male outcome. Standard errors are clustered by Discipline × University. Standard errors are reported in parentheses. Significance levels are defined as follows: $p < 0.1$ *, $p < 0.05$ **, $p < 0.01$ ***.

49%. The early career gender gap on publications is -0.18 standard deviation; the mid career gender gap on publications increases to -0.26 standard deviation. By contrast, we do not detect much difference in impact. If anything, publications by women have a slightly larger impact.

These average gaps mask some significant heterogeneity across fields. The analysis of female representation in Section 3 shows a clear distinction between STEM, where women are still severely underrepresented, and other fields, where women are now well represented. This naturally raises the question of whether gender gaps are worse in STEM. At the intensive margin, since *Publications* and *Impact* are standardized, we can directly compare estimates across fields.

By contrast, comparability of the estimates across fields for *Any Publi* is not immediate, given the differences in baseline levels. We then compute the relative gender gap, equal to the estimated gender gap in *Any Publi* divided by the average likelihood to ever publish among men.

In Humanities and Law and STEM, the relative gender gap at the extensive margin is negative at both career stages and larger in magnitude at mid-career. It is worst at both stages in STEM. By contrast, women are *more* likely to ever publish than men at early career in Biological and Earth Sciences and Social Sciences. This initial advantage is dissipated at mid-career. It remains positive, but statistically non-significant, in Biological and Earth Sciences and becomes negative, and also statistically non-significant, in Social Sciences. Gender gaps at the extensive margin are thus indeed worse in STEM than in the three other fields of research.

By contrast, the gender gap is consistently negative at the intensive margin on publications. Publishing women publish less publications than publishing men in the four fields and at the two career stages. In STEM, Biological and Earth Sciences and Social Sciences, these gender gaps are aggravated at mid-career compared to early career. By contrast, the negative gender gap is slightly lower in magnitude at mid-career in Humanities and Law. Gender gaps at the intensive margin are worst in Biological and Earth Sciences at both career stages. Gender gaps in STEM are very close in magnitude to those in Humanities and Law and Social Sciences at early career and to Social Sciences at mid-career. Economics displays slightly worse gender gaps in publications at both margins than in Social Sciences in general.

Gender gaps in impact are quantitatively modest, always lower in absolute value than 0.05 standard deviation. Publications by women have a higher impact than publications by men in STEM at both career stages. They have a lower impact in Biological and Earth Sciences and Social Sciences, with a non-statistically significant effect at mid-career. Gender gap in impact is statistically not different from zero in Humanities and Law.

Overall, these results demonstrate the presence of strong gender effects in academic trajectories post-graduation and of significant heterogeneity across fields. Of course, raw gender differences could be explained by many factors, such as differential access to good universities or good PhD advisors, motivating the inclusion of controls in the regressions.

4.2 Adjusted gender gaps

We now estimate adjusted gender gaps. We estimate the following model, via OLS, over each field subsample.[5]

$$Y_{iudft} = \beta_{1f} Female_i + \beta_{2f} X_{it} + \mu_{udt} + \epsilon_{iudft} \tag{1}$$

where Y_{iudft} represents research outcome of PhD graduate i, who defended in year t, university u, discipline d and field f. $Female_i$ is a dummy variable equal to 1 if the PhD graduate is female and 0 if he is male. μ_{udt} is a university-discipline-year fixed effect and X_{it} are controls related to supervision.

μ_{udt} is a high-dimensional fixed effect, which captures university-discipline-specific time trends in a non-parametric manner. It controls for time-varying local factors that may affect PhD graduates' academic outcomes such as departments' academic quality and social capital. Controlling for these fixed effects, gender gaps are identified from comparing academic trajectories of women and men who defended their PhD the same year, in the same discipline and university. X_{it} include a dummy equal to 1 if the thesis is in cosupervision, a dummy equal to 1 if a supervisor is female, and a measure of academic productivity of the supervisors.[6]

We also estimate a version of this model with homogeneous parameters ($\beta_{1f} = \beta_1$, $\beta_{2f} = \beta_2$) over the full sample, as well as a version with discipline-specific effects (β_{1d}, β_{2d}) over the Economics subsample.

How are estimated gender gaps affected when controlling for university-discipline-year fixed effects and for supervision characteristics? Estimates are presented in Table 2.

Strikingly, gender gap estimates by field on the likelihood to ever publish are all worse than the raw estimates. Gaps that were positive, in favor of women, are now negative and statistically significant, in Biological and Earth Sciences and Social Sciences. Gaps that were negative are now larger in magnitude, in Humanities and Law and STEM. Gender gap estimates on the number of publications for publishing PhD graduates are also very robust. They remain negative and quantitatively and statistically significant for the 4 fields and 2 career stages. They become larger in magnitude in Humanities and Law and Social Sciences and are little affected in Biological and

[5]Since we include fixed effects at the university-discipline-year level, we remove from the sample observations where only one PhD graduate graduated that year in that university and that discipline. These singleton observations represent about 2% of the overall sample.

[6]For each supervisor s with set of publications at time of defense P_s, we compute the AIS-weighted number of publications at time of defense, equal to $\sum_{p \in P_s} AIS_p$. We then standardize this measure to have standard deviation 1 within discipline. If the PhD is cosupervised, we take the largest of the two supervisors' productivities.

Earth Sciences and STEM. In Economics, gender gaps on publications are worse than the raw gaps, and are negative and statistically significant at the extensive and intensive margins and at early and mid-career. Gender gap estimates on impact remain, overall, quantitatively modest and also tend to worsen. In particular, gender gaps on impact in STEM at both career stages, that were positive before, are now close to zero and statistically non-significant. Gender gaps on impact in Humanities and Law and Social Sciences are also non-significant, while gender gaps in Biological and Earth Sciences at both career stages are significantly negative, even though quantitatively modest, below 0.05 standard deviation in magnitude.

This shows that a significant part of the raw gender gaps captures correlation between gender and the controls. A worsening of the gender gaps when adjusting is a sign of *positive* female sorting, i.e., women being overrepresented in the university-discipline-years with better research outcomes. Such positive sorting happens, for instance, when women are better represented among PhD graduates of higher-quality departments.

How do adjusted gender gaps in STEM compare to those in other fields? At mid-career, the relative gender gap at the extensive margin is still worst in STEM than in the other fields. At early career, it is now the second worst, behind Humanities and Law.[7] At the early career intensive margin, by contrast, gender gaps in STEM are now lowest in magnitude among the 4 fields. At mid-career, the intensive margin gender gap in STEM is lower in magnitude than in Biological and Earth Sciences and in Social Sciences.

Overall, the estimations reveal a nuanced picture. The worst representation of women in STEM appears to be associated with worse gender gaps at the extensive margin, but not at the intensive margin. Women indeed drop out of academia more in STEM. However, conditional on publishing, gender gaps in terms of number of publications and impacts are not worse in STEM.

To sum up, accounting for university-discipline-year fixed effects and supervision controls, we find that women are less likely to ever publish than men and that publishing women publish less than publishing men in the 4 fields of research and 2 career stages. Gender gap estimates are generally larger in magnitude than in the raw data, consistent with positive female sorting in space, discipline and time. Gender gaps in STEM, while bad, are not particularly worse than in the other fields of research.

[7] The absolute gender gap at the extensive margin is worst in STEM than in the other three fields at both career stages and for both raw and adjusted gaps.

Table 2: Adjusted gender gaps

	All Fields	Humanities and Law	Biological and Earth Sc.	STEM	Social Sciences	Economics
			Before t+4			
Any Publi	**-0.0496*****	**-0.0296*****	**-0.0319*****	**-0.0833*****	**-0.0195*****	**-0.0387*****
	(0.00261)	(0.00391)	(0.00539)	(0.00389)	(0.00448)	(0.0103)
Av. Male Any Publi	0.492	0.153	0.481	0.679	0.244	0.278
Relative Gender Gap	-10%	-19%	-7%	-12%	-8%	-14%
# PhD Graduates	244754	50160	43643	107986	42965	8162
if Any Publi = 1						
Publications	**-0.254*****	**-0.371*****	**-0.310*****	**-0.217*****	**-0.267*****	**-0.366*****
	(0.00956)	(0.0585)	(0.0210)	(0.00937)	(0.0362)	(0.0660)
Impact	**-0.0129**	**0.0153**	**-0.0474*****	**-0.000195**	**-0.0317**	**-0.0102**
	(0.00800)	(0.0316)	(0.0150)	(0.0105)	(0.0235)	(0.0572)
# PhD graduates *if Any Publi = 1*	107726	6236	22033	70199	9258	1955
			Between t+5 and t+10			
Any Publi	**-0.0758*****	**-0.0306*****	**-0.0806*****	**-0.119*****	**-0.0315*****	**-0.0500*****
	(0.00360)	(0.00464)	(0.00874)	(0.00409)	(0.00576)	(0.0123)
Av. Male Any Publi	0.380	0.155	0.401	0.493	0.233	0.264
Relative Gender Gap	-20%	-20%	-20%	-24%	-14%	-19%
# PhD graduates	187391	38954	34168	81998	32271	6340
if Any Publi = 1						
Publications	**-0.348*****	**-0.229*****	**-0.426*****	**-0.318*****	**-0.373*****	**-0.434*****
	(0.0131)	(0.0467)	(0.0228)	(0.0145)	(0.0380)	(0.0644)
Impact	**-0.00931**	**-0.0107**	**-0.0385****	**0.00921**	**-0.0222**	**-0.0410**
	(0.00912)	(0.0265)	(0.0149)	(0.0138)	(0.0271)	(0.0868)
# PhD graduates *if Any Publi = 1*	60947	4789	13400	36594	6164	1373

Notes: The table reports adjusted gender gaps in publication outcomes for PhD graduates, estimated separately for two time windows: before t+4 and between t+5 and t+10, where t denotes the year of PhD graduation. Any Publi is a binary variable equal to 1 if the PhD graduate has at least one publication in the given period. Publications and Impact are continuous outcomes measured conditional on having at least one publication. All specifications include Discipline × Year × University fixed effects, as well as controls for co-supervision status, supervisor's publication quality, and supervisor gender. The Relative Gender Gap is computed as the ratio of the female coefficient to the average male outcome, calculated on the estimation sample. The sample is slightly restricted relative to Table 1 (by approximately 5%) due to the removal of singleton observations arising from the fixed effects structure. Standard errors are clustered by Discipline × University and reported in parentheses. Significance levels are defined as follows: $p < 0.1$ *, $p < 0.05$ **, $p < 0.01$ ***.

4.3 Evolution of gender gaps over time

Since female representation has increased over time in all fields, a natural question is whether the gender gaps have also evolved. To analyze gender gap evolution, we consider separate periods of 5 years and rerun separate regressions for each of these periods, including university-discipline-year fixed effects and supervision controls. More precisely, the 1988 - 2012 time window of the middle career sample is partitioned in five 5-year periods and there is an additional 6-year period in the partition of the 1988 - 2018 time window of the early career sample. Figure 2 depicts how estimated gender gaps in *Any Publi* and in *Publications* vary over time for the four fields and two career stages. Figure C2 in the Online Appendix depicts similar graphs for *Impact*.

The results reveal a limited evolution of gender gap estimates over time. French academia is clearly not converging towards gender parity in academic trajectories post-graduation. At the extensive margin, we see a slight worsening of the gender gap in the 1990's, when female representation was increasing rapidly, followed by a slight improvement in the most recent periods. In STEM, the early career gender gap in 2013 - 2018 is lowest in magnitude compared to the other periods, although still negative and quantitatively and statistically significant. In Social Sciences, the early career gender gap in 2013 - 2018 is statistically not different from zero. However, it was also not statistically different from zero in 1988 - 1992, and then became significantly negative in the 4 periods in-between. In Biological and Earth Sciences, the early career gender gap is negative throughout, although lower in magnitude in recent periods and statistically not different from zero in 2013 - 2018. In Humanities and Law, the early career gender gap is lowest in magnitude in the first two periods, although everywhere statistically different from zero. At the extensive marginGender gaps at mid-career display similar field-specific patterns, while being generally negative and larger in magnitude than at early career.

The gender gaps on publications and impacts at the intensive margin appear to vary little over time.[8] As in the overall sample, gender gaps on publications are generally large and negative while gender gaps on impact are quantitatively small. Overall, this shows that the large increase in female representation among PhD graduates over the period is not associated with a reduction in gender differentials in academic trajectories post-graduation.

[8]Estimates for Humanities and Law in the first period, 1988 - 1992, are very imprecise, due to a small number of publishing PhD graduates in this field and period. We removed them from the graphs for clarity.

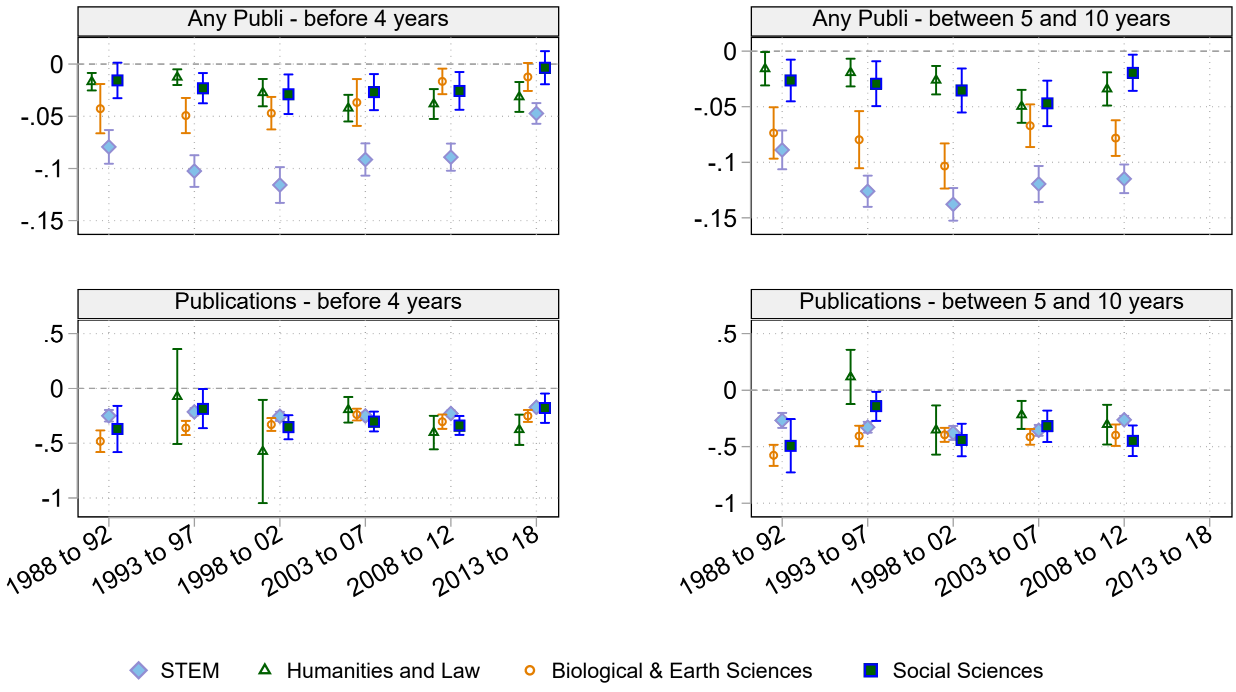

Figure 2: Evolution of gender gaps over time

5 Missing women

In this section, we compute the number of "missing women in academia". We focus on the mid-career sample, formed of PhD graduates in French universities in all disciplines (except Health and Medical Sciences) who graduated between 1988 and 2012. Our aim is to compute the number of women from this sample who "should" have been academically productive at mid-career but were not.

Any computation of missing women relies on counterfactual scenarios, as discussed in the literature on missing women in poor countries (e.g. Sen (1990), Coale (1991), Anderson and Ray (2010)). Here, we build these counterfactual scenarios from the estimates of model (1) for *Any Publi* described in the previous section. Consider a female PhD graduate i who defended in university u, discipline d and year t and with supervision controls X_{it}. From the estimated model, we can obtain two probabilities. First, the estimated probability that i publishes at least one publication between 5 and 10 years post-graduation, $\hat{p}_i(Female_i = 1)$. And second, the predicted probability of publishing if i had been a male PhD graduate defending in the same university, discipline, and year and with the same supervision controls, $\hat{p}_i(Female_i = 0)$.

The estimated number of missing women m_{Wf} in a field is then the sum of the differences in probabilities over all female PhD graduates from the field: $m_{Wf} = \sum_i \hat{p}_i(Female_i = 0) - \hat{p}_i(Female_i = 1)$. Given our linear specification, this is equal to the estimated gender gap at the extensive margin, $-\hat{\beta}_{1f}$, times the number of female PhD graduates in the field, n_{Wf}.

$$m_{Wf} = -\hat{\beta}_{1f} n_{Wf}$$

We then sum these numbers over the four fields to obtain an estimate of the overall number of missing women, $m_f = \sum_f m_{Wf}$. Similarly, we can compute the number of missing publications by publishing women, m_{Pf}, by summing over all publishing women in a field the difference in *Publications* when setting $Female_i = 0$ instead of $Female_i = 1$. This is equal to the estimated gender gap at the extensive margin times the number of publishing women.

Table 3 provides estimates of the number of missing women in academia and of the number of missing publications by publishing women.

Table 3: Missing Women and Missing Papers

	Missing Women				Missing Papers			
Fields	n_W	$-\hat{\beta}_1$ (Any publi)	m_W	Conf. Inter. 95% (m_W)	$n_{W\|P=1}$	$-\hat{\beta}_1$ (# publi)	m_P	Conf. Inter. 95% (m_P)
Humanities and Law	19,258	0.0306	589	[414; 764]	2,603	0.229	596	[358; 834]
Biological and Earth Sc.	16,837	0.0806	1,357	[1,069; 1,645]	6,555	0.426	2,792	[2,499; 3,085]
STEM	23,748	0.119	2,826	[2,635; 3,017]	8,912	0.318	2,834	[2,580; 3,088]
Social Sc.	15,841	0.0315	499	[320; 678]	3,535	0.373	1,318	[1,055; 1,581]
All Fields	75,684	0.0758	5,737	[5,203; 6,271]	21,605	0.348	7,519	[6,964; 8,074]

n_W is the number of female PhD graduates in each field. $n_{W|P=1}$ is the number of female PhD graduates who published at least once between $t+5$ and $t+10$. $-\hat{\beta}_1$ is the estimated gender gap in publication outcomes from equation 1, where *Any Publi* refers to a binary indicator equal to 1 if the individual published at least one paper, and *# Publi* refers to the number of publications conditional on publishing. $m_W = n_W \times (-\hat{\beta}_1)$ and $m_P = n_{W|P=1} \times (-\hat{\beta}_1)$ represent the estimated number of missing women and missing papers, respectively. 95% confidence intervals are computed as $m \pm 1.96 \times n \times SE$, where SE is the clustered standard error of $\hat{\beta}_1$. Standard errors are clustered by Discipline $\times$ University.

Overall, we estimate that in the 25-year period from 1988 to 2012, 5,737 female PhD graduates out of the 54,079 who did not publish at mid-career should have been academically active. This represents a 27% increase of the population of publishing female PhD graduates and 7% of the full population of female PhD graduates. The bulk of these numbers is coming from STEM, both because of a direct size effect - STEM is a larger field and has more female PhD graduates than the other fields, despite the fact that women are less well-represented in STEM - and of the fact that the estimated negative gender gap is larger in STEM.

Note that this methodology has several limitations. First, these computations are performed conditionally on the actual population of PhD graduates. This mechanically reduces the number of missing women in fields with a lower proportion of female PhD graduates. Accounting for gender imbalances in the PhD population is an important direction for future research - and would increase the estimated number of missing women in academia given their persistent underrepresentation in STEM. Second, these computations rely on the assumption that there is no predetermined limit on the number of PhD graduates in a field and year who can end up being academically active. An alternative scenario would be to assume that this number is fixed - which would decrease the estimated number of missing women.

6 Conclusion

In this paper, we provide the first comprehensive, country-wide analysis of gender gaps in academic trajectories after PhD graduation, covering 30 years and all academic disciplines. Female representation among PhD graduates has improved over time in all fields. While women remain severely underrepresented in STEM, they are now slightly overrepresented in non-STEM fields. Are gender differentials in academic trajectories, then, less pronounced in non-STEM fields and did the evolution of these differentials track the evolution of female representation?

We document sizable and persistent negative gender gaps in academic trajectories. Controlling for university-discipline-year fixed effects and supervision characteristics, we find that women are less likely to ever publish than men and that publishing women publish less than publishing men in all fields and career stages. The effects are quantitatively large, up to -24% relative to men at the extensive margin and -0.43 standard deviation at the intensive margin. Unconditional estimates appear less negative and even show positive gaps, in favor of women, in Biological and Earth Sciences and in Social Sciences at the extensive margin, indicating positive sorting of women across space, discipline and time. These negative gaps have not declined over time. They tend to be worse in STEM at the extensive margin, but not at the intensive margin. Women PhD graduates indeed drop out of academia more in STEM. However, gender gaps on number of publications for publishing women are not worse in STEM.

We also assess gender gaps in publications' impacts, measured by the average Article Influence Score across all publications of the PhD graduate. Estimated effects are quantitatively small and precisely estimated. In our preferred specification, we cannot reject equality in average publication impact between men and women in Humanities and Law, Social Sciences and STEM. Average impact is significantly lower for women in Biological and Earth Sciences, with a gender gap lower than -0.05 standard deviation.

Overall, our results indicate the presence of substantial and persistent gendered barriers in the critical early years following PhD completion. These barriers reduce the likelihood that women establish successful academic careers compared to men. We estimate that removing them would increase the population of publishing female PhD graduates by about 27%. Moreover, our analysis suggests that these barriers are largely disconnected from those operating during graduate studies. Reaching, or even exceeding, gender parity among PhD graduates does not automatically translate into lower gender differentials in academic trajectories. It would be interesting, in future research, to try and identify the precise causes behind these gender gaps.

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A Data theses.fr - Detailed Procedure

We construct our dataset using data from *theses.fr*, which provides records of all PhD theses defended in French universities between 1988 and 2021. *theses.fr* is a centralized public platform that systematically compiles data from university catalogs across France, sourced through library and documentation services within higher education and research institutions, establishing it as the most comprehensive and reliable platform for French PhD graduation.

The dataset is not immune to limitations. Data entry occurs manually at various stages, which introduces the potential for spelling inconsistencies. Furthermore, certain theses may go unreported due to a lack of submission by graduates, loss, or failure to meet quality control standards, which we estimate affects approximately 5% of theses each year. In addition, the processing of records is time-intensive, making the data for 2022 potentially incomplete. Additionally, an observed scarcity of records prior to 1988 suggests further underreporting. Consequently, we restrict our sample to the period from 1988 to 2021.

From an initial sample of 407,260 theses recorded between 1988 and 2021, we impose a series of exclusions to ensure data reliability. Theses supervised by more than two advisors-constituting roughly 2% of the dataset-are excluded, yielding a refined dataset of 397,536 observations. Additional filters are applied to exclude records with incomplete names for PhD candidates or supervisors, as well as cases with missing discipline information, resulting in a final dataset of 393,259 theses. At this stage, we exclude theses in medicine due to reliability concerns, which we discuss in detail in Section A.2, leaving a total of 335,796 observations.

For each thesis, we gathered information on the research discipline, defense year, university affiliation, and full names of the PhD student and supervisor(s). In the sections that follow, we detail the data-cleaning procedures applied to discipline and university affiliation, explain the exclusion of health and medical sciences, and outline our methodology for associating gender with first names.

A.1 Gender association

In this study, we determine the gender of both PhD students and supervisors based on first names. Our primary source is the INSEE database, which compiles first names assigned in France from 1900 to 2020, including the gender distribution for each name over the period 1940–2020. We focus on this range, assuming that the majority of PhD students in our dataset were born after 1940. For names associated with both genders, we establish a reliable gender ratio and retain only those names where one gender represents at least 95% of total occurrences; names below this threshold are treated as indeterminate. This process allows us to identify the gender for 305,187 out of 335,796 PhD student first names. Recognizing the limitations posed by foreign names, we supplement INSEE data with governmental databases from Australia, Canada, Spain, Sweden, the UK, and the US.

Through additional data collection from these international sources, we resolve the gender of an additional 9,246 PhD students. We further employ the methodology of Benveniste (2023), which classifies names based on the last two letters and the associated gender probability, allowing us to identify the gender of 3,004 more PhD students. In total, we successfully identify the gender of 313,425 doctoral students, covering 93% of the sample. Of the remaining 7%, 3% (8,166 names) represent names used by both genders without a clear distributional majority (e.g., Camille, Claude). Using the same approach, we successfully associate a gender for 95% of PhD supervisors.

A.2 Disciplines

The categorization of discipline fields in *theses.fr* is imprecise, partly due to manual data entry. The database originally contained around 22,000 unique entries for the discipline variable, which we grouped into twenty-two subcategories and further into four broader categories based on the Australian and New Zealand Standard Research Classification (ANZSRC). To classify these entries, we adopted a keyword-based approach, manually associating each entry with relevant discipline categories. We began by filtering with specific keywords unique to each category, as illustrated in the following examples:

Example

"CHIMIE ORGANIQUE" for "Chemical Sciences"

"INFORMATIQUE" for "Information, computing and Communication Sciences"

"SCIENCES BIOLOGIQUES" for "Biological Sciences" ...

Following this, we applied progressively broader keywords, carefully verifying that each association was accurate to avoid misclassification. For example, general keywords like "MAGNETISME", "LANGUES," and "VEGETAL" were used, corresponding to "Physical Sciences", "Language and Culture," and "Biological Sciences", respectively.

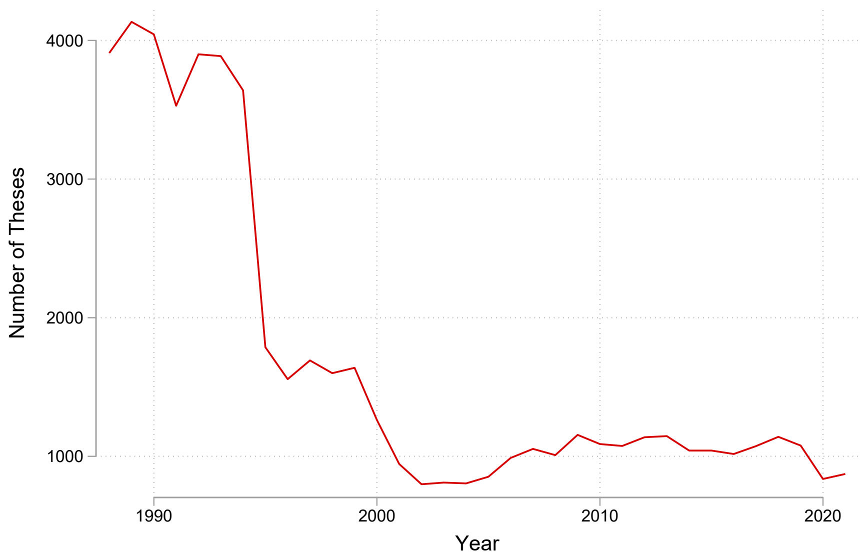

Figure A1: Number of Thesis by Year of Defense in Health and Medical Sciences

In cases of ambiguous or unknown disciplines, we examined thesis titles and applied the same keyword methodology. Despite these efforts, discipline association may still contain errors, especially for multidisciplinary theses that we must assign to a single category. To account for this, we created four overarching categories to group similar subjects: Humanities and Law, Biological and Earth Sciences, Sciences, Technology and Engineering, and Social Sciences.

Drop *Health and Medical Sciences* discipline. In this section, we discuss the unreliability of Health and Medical Sciences thesis data prior to the 2000s. Our analysis identified notable irregularities in medical theses data, particularly around 1994. We traced the origin of these discrepancies to the data selection mechanism in *theses.fr*, which automatically selects defended doctoral theses and excludes documents not categorized as such. However, in the French health sciences domain, "*Thèses d'exercice*" - theses defended to obtain a State Diploma of Doctor required for medical practice-are often included. These are distinct from doctoral theses intended to confer the national diploma of doctor (*diplôme national de doctorat*). Unfortunately, during data import into *theses.fr*, a substantial number of *Theses d'exercice* were incorrectly labeled as doctoral theses, introducing bias.

Figure A1 displays the number of theses defended in health and medical sciences since 1988, showing that institutions began systematically distinguishing between doctoral theses and *Thèses d'exercice* around the early 2000s. As we aim to focus on theses from before 2000, we must exclude medical theses from our sample to avoid biasing our study.

A.3 University

In recent years, French universities have been undergoing a series of institutional mergers, intended to enhance their international visibility and competitiveness[9]. To ensure consistency in our analysis, we standardized university codes following the documentation provided by theses.fr[10] and tracked changes in institutional names over time. Between 2007 and 2020, 26 new universities were established through the consolidation of 76 existing institutions. For example, in 2013, Aix-Marseille University was formed by merging Aix-Marseille 1, Aix-Marseille 2, and Aix-Marseille 3.

In certain cases, however, institutions have subsequently split, complicating the distinction between former codes. In such instances, it is more practical to apply a single code for universities that have separated, even at the cost of some specificity. For example, the University of Paris-Saclay was initially formed in 2015 as a merger of 11 institutions, only to divide into two distinct entities by the end of 2019.

Table A1, A2, and A3 provide a detailed list of all universities and their coding changes, while Table A4 covers the National Institutes of Polytechnics, and Table A5 presents the Higher Education Establishments. Each institution is listed with its associated code and any historical coding changes from 1988 to 2021. Any changes or codes appearing before or after this period are not documented. A blank description indicates no changes during the specified timeframe.

[9] https://www.enseignementsup-recherche.gouv.fr/fr/premier-bilan-des-fusions-d-universites-realisees-entre-2009-et-2017-47515

[10] https://documentation.abes.fr/guide/html/regles/CodesUnivEtab.htm

Code	University	Description
AGUY+ANTI+YANE*	Antilles-Guyane	ANTI and YANE since 2015
AIX1	Aix-Marseille 1	See AIXM since 2012
AIX2	Aix-Marseille 2	See AIXM since 2012
AIX3	Aix-Marseille 3	See AIXM since 2012
AIXM	Aix-Marseille	Creation 2012
AMIE	Amiens	
ANGE	Angers	
ANTI	Antilles	Creation 2015
ARTO	Artois	
AVIG	Avignon	
AZUR (=COAZ)**	Univ. Côte d'Azur (ComUE)	Creation 2016, changing code in 2020
BELF	Belfort Montbéliard	See UBFC since 2017
BESA	Besançon	See UBFC since 2017
BOR1 + BOR4***	Bordeaux 1 + 4	See BORD since 2014
BOR2	Bordeaux 2	See BORD since 2014
BOR3	Bordeaux 3	See BORD since 2014
BORD	Bordeaux	Creation 2014
BRES	Brest - Bretagne occidentale	
CAEN	Caen	See NORM since 2017
CERG (=CYUN)	Cergy-Pontoise	Changing code CYUN in 2020
CHAM	Chambéry	See GREN since 2010
CLF1	Clermont-Ferrand 1	See CLFA since 2021
CLF2	Clermont-Ferrand 2	See CLFA since 2021
CLFA (=UCFA)	Univ. Clermont Auvergne	Changing code UCFA in 2020
COMP	Compiègne	
CORT	Corte	
DIJO	Dijon	See UBFC since 2017
DUNK	Littoral Dunkerque	
EVRY	Evry Val d'Essonne	See SACL since 2015
GRAL	Univ. Grenoble Alpes	
GRE1	Grenoble 1	See GREN since 2010
GRE2	Grenoble 2	See GREN since 2010
GRE3	Grenoble 3	See GREN since 2010
GREN(=GREA = GRAL)	Grenoble	Changing code in 2015, 2020
LARE	La Réunion	
LARO	La Rochelle	
LEHA	Le Havre	See NORM since 2017
LEMA	Le Mans	

Table A1: Universities

All the code of the universities associated with their name and the evolution of their code over the years. We focus on the period 1988 to 2021, any changes and code that appears before or after are taken into account. If the description is empty, it means that there is no change during the period. * Guyane and Antilles were part of the same university at the beginning and then split, so we have to do only one university with all(because we don't know who was in which university); ** The sign equal, when the code name changed but represents the same university; *** BOR4 since 1995 for Law, Social Sciences and politics, Economics and Management theses), so we have to merge the two universities

Code	University	Description
LIL1	Lille 1	See LILU since 2018
LIL2	Lille 2	See LILU since 2018
LIL3	Lille 3	See LILU since 2018
LILU	Univ.polfLille	Creation 2018
LIMO	Limoges	
LORI	Lorient-Bretagne sud	
LORR	Univ. de Lorraine	Creation 2012
LYO1	Lyon 1	See LYSE since 2015
LYO2	Lyon 2	See LYSE since 2015
LYO3	Lyon 3	See LYSE since 2015
LYSE	Lyon (COMUE)	Creation 2015
MARN	Marne la Vallée	See PEST since 2008
METZ	Metz	See LORR since 2012
MON1	Montpellier 1	See MONT since 2015
MON2	Montpellier 2	See MONT since 2015
MON3	Montpellier 3	
MONT	Montpellier	Creation 2015
MULH	Mulhouse	
NAN1	Nancy 1	See LORR since 2012
NAN2	Nancy 2	See LORR since 2012
NANT	Nantes	
NCAL	Nouvelle Calédonie	
NICE	Nice	See AZUR since 2016
NIME	Nîmes	
NORM	Normandie (COMUE)	Creation 2017
PA01	Paris 1	
PA02	Paris 2	
PA03	Paris 3	See USPC de 2015 à 2019
PA04	Paris 4	See SORU since 2018
PA05	Paris 5	See USPC de 2015 à 2019 See UNIP since 2019
PA06	Paris 6	See SORU since 2018
PA07	Paris 7	See USPC de 2015 à 2019 See UNIP since 2019
PA08	Paris 8	
PA09	Paris 9	See PSLE since 2016
PA10	Paris 10	
PA11	Paris 11	See SACL since 2015
PA12	Paris 12	See PEST de 2008 à 2020
PA13	Paris 13	See USPC de 2015 à 2019
PACI+NCAL+POLF*	Pacifique	NCAL and POLF since 1999
PAUU	Pau	
PERP	Perpignan	
PEST(=PESC)**	Paris Est (COMUE)	
POIT	Poitiers	
POLF	Polynésie française	

Table A2: Universities

All the code of the universities associated with their name and the evolution of their code over the years. We focus on the period 1988 to 2021, any changes and code that appears before or after are taken into account. If the description is empty, it means that there is no change during the period. * Nouvelle Calédonie and Polynésie française were part of the same university at the beginning and then split, so we have to use only one code with both as we can't distinguish them. ** PEST changed its name in 2015 to PESC

Code	University	Description
REIM	Reims	
REN1	Rennes 1	
REN2	Rennes 2	
ROUE	Rouen	
SACL+UPAS+IPPA+IAVF*	Univ. Paris-Saclay (ComUE)	Creation in 2015
SORU	Sorbonne Univ.	
STET	Saint-Etienne	See LYSE since 2015
STR1	Strasbourg 1	See STRA since 2009
STR2	Strasbourg 2	See STRA since 2009
STR3	Strasbourg 3	See STRA since 2009
STRA	Strasbourg	Creation 2009
TOU1	Toulouse 1	
TOU2	Toulouse 2	
TOU3	Toulouse 3-Ec. nationale vétérinaire	
TOUL	Toulon	
TOUR	Tours	
TROY	Troyes	
UBFC	Bourgogne Franche-Comté	Creation 2017
UCFA	Univ. Clermont-Auvergne	
UEFL	Univ. Gustave Eiffel	
UNIP	Univ. de Paris	Creation 2019
UPHF	Univ. Polytech. Hauts-de-France - Valenciennes	
USPC+PA03+PA13 +INAL+UNIP**	Sorbonne Paris Cité	Creation in 2019
VALE	Valenciennes	See UPHF since 2019
VERS	Versailles St Quentin en Yvelines	See SACL since 2015
YANE	Guyane	Creation 2015

Table A3: Universities
All the code of the universities associated with their name and the evolution of their code over the years. We focus on the period 1988 to 2021, any changes and code that appears before or after are taken into account. If the description is empty, it means that there is no change during the period. * IAVF is a new branch in 2016 and SACL was divided into UPAS and IPPA in 2019, as we can't distinguish, we use the same code for the three. ** There is a merge and then a split of universities, so we use one code for PA03, PA13, INAL, and UNIP only after 2019.

Code	Institute	Description
INPG	Institut national polytechnique - Grenoble	See GREN since 2009
INPL	Institut national polytechnique - Lorraine	
INPT	Institut national polytechnique - Toulouse	
IPPA	Institut Polytechnique de Paris	

Table A4: National Institute of Polytechnics
All the code of the universities associated with their name and the evolution of their code over the years. We focus on the period 1988 to 2021, any changes and code that appears before or after are taken into account. If the description is empty, it means that there is no change during the period.

Code	Establishment	Description
AGPT+EIAA +ENGR+INAP*	AgroParisTech	See SACL since 2015
CLIL	Centrale Lille Institut	
CNAM	Conservatoire national des arts et métiers	
CSUP	CentraleSupélec	See SACL since 2015
DENS	Ec. normale supérieure - Cachan	See SACL since 2015
ECAP	Ec. centrale des arts et manufactures de Paris	See SACL since 2015
ECDL	Ec. centrale de Lyon	See LYSE since 2015
ECDM	Ec. centrale de Marseille	
ECDN	Ec. centrale de Nantes	See CLIL since 2020
ECLI	Ec. centrale de Lille	See CLIL since 2020
EHEC	Ec. des hautes études commerciales	See SACL since 2015
EHES	Ec. des hautes études en sciences sociales	
EIAA	Ec. nationale supérieure des industries alimentaires - Massy	See AGPT since 2007-
EMAC	Ec. nationale des Mines d'Albi-Carmaux	
EMAL	IMT Mines Alès	
EMNA	Ec. des Mines de Nantes	See IMTA since 2017
EMSE	Ec. nationale supérieure des Mines - Saint-Etienne	
ENAM	Ec. nationale supérieure d'arts et métiers	See HESA since 2020
ENCM	Ec. nationale supérieure de chimie de Montpellier	
ENCP	Ec. nationale des chartes	
ENCR	Ec. nationale supérieure de chimie de Rennes	
ENGR	Ec. nationale du génie rural, des eaux et forêts	See AGPT since 2007
ENIB	Ec. nationale d'ingénieurs de Brest	
ENIS	Ec. nationale d'ingénieurs de Saint-Etienne	See LYSE since 2015
ENMP	Ec. nationale supérieure des Mines - Paris	See PSLE since 2016
ENPC	Ec. nationale des ponts et chaussées	See PEST since 2008
ENSL	Ec. normale supérieure (sciences) - Lyon	See LYSE since 2015
ENSR	Ec. normale supérieure de Rennes	
ENST	Ec. nationale supérieure des télécommunications	See SACL since 2015
ENSU	Ec. normale supérieure- Paris (rue d'Ulm)	See PSLE since 2016
ENTA	Ec. nationale supérieure de techniques avancées Bretagne	
ENTP	Ec. nationale des travaux publics	See LYSE since 2015
EPHE	Ec. pratique des hautes études	See PSLE since 2016
EPXX	Ec. polytechnique	See SACL since 2015
ESAE	ISAE	
ESEC	Ec. supérieure des sciences économiques et commerciales	
ESMA	Ec. nationale supérieure de mécanique et d'aérotechnique	
ESTA	Ec. nationale supérieure de techniques avancées	See SACL since 2015
GLOB	Institut de physique du Globe	See USPC since 2015
HESA	HESAM	
IAVF	Institut agronomique, vétérinaire et forestier de France - Paris	
IEPP	Institut d'études politiques - Paris	
IMTA	Ec. nationale supérieure Mines-Télécom Atlantique Bretagne Pays de la Loire	
INAL	Institut national des langues et civilisations orientales (INALCO)	See USPC since 2015
INAP	Institut national d'agronomie - Paris Grignon	See AGPT since 2007
IOTA	Institut d'optique théorique et appliquée - Palaiseau	SACL UPAS
ISAB	Institut national des sciences appliquées Val de Loire - Bourges	
ISAL	Institut national des sciences appliquées - Lyon	See LYSE since 2015
ISAM	Institut national des sciences appliquées - Rouen	See NORM since 2017
ISAR	Institut national des sciences appliquées - Rennes	
ISAT	Institut national des sciences appliquées - Toulouse	
MNHN	Museum d'histoire naturelle	
MTLD	Ec. nationale supérieure Mines-Télécom Lille Douai	
NSAI	Ec. nationale de la Statistique et de l'Analyse de l'Information - Rennes	
NSAM	SupAgro - Montpellier	
NSAR	Agrocampus Ouest - Rennes	
OBSP	Observatoire de Paris	See PSLE since 2016
ONIR	Ec. nationale vétérinaire - Nantes	
ORLE		
PSLE(=UPSL)	Paris Sciences et Lettres (ComUE)	Creation 2016
TELB	Ec. nationale supérieure des TelecompolfBretagne - Brest	See IMTA since 2017
TELE	Institut national des télécommunications	See SACL since 2015

Table A5: Higher Education Establishment

All the code of the universities associated with their name and the evolution of their code over the years. We focus on the period 1988 to 2021, any changes and code that appears before or after are taken into account. If the description is empty, it means that there is no change during the period. * EIAA+ENGR+INAP merged to become AGPT in 2007 we use one code for the three. ** Change code in 2020

B Data Scopus - Detailed Procedure

This section describes the procedure used to match PhD graduates and their supervisors to Scopus author profiles and to construct publication-based productivity measures. The matching pipeline proceeds in two parallel streams - one for PhD graduates and one for supervisors - each following the same broad logic: an initial name-based query to Scopus, followed by successive filters based on research field, name similarity, publication history, and manual review, culminating in the consolidation of multiple Scopus IDs into a single profile per individual. The two streams differ in one important respect: given that excluding a supervisor from the sample entails dropping all PhD graduates under their supervision, the supervisor pipeline relies more heavily on manual review and applies more conservative exclusion thresholds, so as to minimise sample attrition.

The final analytical sample, described in Section B.5, is obtained by merging the datasets from *theses.fr* with publication-level information retrieved from Scopus and journal quality indicators from Clarivate.

B.1 PhD Students

B.1.1 Retrieving of "Authors IDs"

We submitted a query to the Scopus platform using the *name* and *surname* of each PhD graduate recorded in theses.fr. The initial request covered 335,796 unique PhD graduates. Due to retrieval errors on the Scopus platform, 26 entries could not be returned, of which 24 were identified as mis-retrieved at the download stage and an additional 2 were lost during cleaning, leaving a working sample of 335,770 PhD graduates. Note that this working sample includes PhD graduates and supervisors for whom gender could not be determined; these observations are retained throughout the matching pipeline and only excluded at the final sample construction stage (Section B.5).

For each graduate, the Scopus query returned, where available: a name and surname, one or more *Author IDs*, an institutional affiliation, and up to three primary research fields.

After this first retrieval, PhD graduates are distributed across four categories:

Count	**Description**
150,044	One unique Scopus ID matched.

114,719	No Scopus ID found.
62,473	Between 2 and 23 possible Scopus IDs (ambiguous match).
8,534	Query returned more than 23 results (name too common).
335,770	Total PhD graduates in working sample.

Field-based selection (Step 1a). Each Scopus candidate profile carries up to three subject-area codes (e.g. `MEDI`, `ENGI`, `ARTS`), which we recode into broader disciplinary buckets and compare against the thesis discipline from theses.fr. A PhD–Scopus pair is retained as a field match if at least one Scopus bucket aligns with the thesis field. Psychology (`PSYC`) is treated as ambiguous (mapping to either `ARTS` or `MEDI`) and is deprioritised when other fields are available. When a PhD has multiple candidate profiles, non-matching ones are dropped; if none match, one row is retained with the profile flagged as unresolved. Table B7 describes the full field-equivalence mapping between theses.fr and Scopus disciplines.

After this filter, the distribution becomes:

Count	**Description**
147,239	One unique, field-validated Scopus ID.
146,356	No Scopus ID found (including profiles excluded by field mismatch).
33,641	Still ambiguous (multiple field-compatible profiles remain).
8,534	Name too common; excluded from further matching.
335,770	Total.

Name-similarity filter (Step 1b). Building on the field-based selection, we apply fuzzy string matching to further reduce ambiguous cases. Surnames and given names from both theses.fr and Scopus are standardised (lowercase, diacritics removed) before computing similarity scores using the *matchit* package in Stata. A threshold of 0.40 is applied: calibrated by manual inspection of name pairs across the score distribution, this value corresponds to a clear qualitative break, below which pairs consistently belong to different surnames, and above which they represent

Table B7: All Disciplines in Scopus and their equivalent in Thèses.fr

Disciplines SCOPUS	Abbrev. Scopus	Disciplines Thèses.fr
Arts and Humanities	**ARTS**	Philosophy and Religion Language and Culture Journalism, Librarianship and Curatorial Studies The Arts Law, Justice and Law Enforcement History and Archaeology Architecture, Urban Environment and Building
Social Sciences	**SOCI**	Behavioural and Cognitive Sciences Studies in Human Society Education Policy and Political Science
Economics, Econometrics and Finance	**ECON**	Economics
Business, Management and Accounting Decision Sciences	**BUSI** DECI	Commerce, Management, Tourism and Services
Chemical Engineering Chemistry	**CENG** CHEM	Chemical Sciences
Computer Science	**COMP**	Information, computing and Communication Sciences
Biochemistry, Genetics and Molecular Biology	**BIOC**	Biological Sciences
Physics and Astronomy Materials Science	**PHYS** MATE	Physical Sciences
Engineering Energy Multidisciplinary	**ENGI** ENER MULT	Engineering and technology
Environmental Science Agricultural and Biological Sciences Veterinary	**ENVI** AGRI VETE	Agricultural, Veterinary and Environmental Sciences
Earth and Planetary Sciences	**EART**	Earth Sciences
Mathematics	**MATH**	Mathematical Sciences
Medicine Immunology and Microbiology Health Professions Pharmacology, Toxicology and Pharmaceutics Nursing Dentistry Psychology	**MEDI** IMMU HEAL PHAR NURS DENT PSYC	Medical and Health Sciences

plausible variants of the same name. An additional constraint is imposed on given names: the first letter must match. When multiple Scopus candidates remain for the same PhD, those failing the name or surname filter are dropped, retaining the best-scoring candidate.

After this step:

Count	**Description**
148,699	One unique Scopus ID (field- and name-validated).
146,335	No Scopus ID found.
32,202	Still ambiguous (multiple candidates remain after filtering).
8,534	Name too common; excluded.
335,770	Total.

In total, 178,310 PhD graduates - those with at least one field- and name-validated candidate (148,699 + 32,202 − 2 lost to retrieval errors at the next step) - are forwarded to Step 2 for article retrieval. The remaining observations (no Scopus profile, common names, or unresolved mismatches) do not require publication retrieval.

B.1.2 Step 2 - Retrieval of Publications

We retrieved the full publication history for all 178,310 PhD graduates with at least one validated Scopus candidate. This represents approximately 4.5 million publications in total, which are used to detect remaining mismatches. Two additional PhD graduates could not be retrieved at this stage (mis-retrieval errors); PhD graduates with a valid Scopus ID but zero retrieved publications are recorded with a publication count of zero.

After retrieval, the distribution is:

Count	**Description**
148,717	One unique Scopus ID (field- and name-validated).
146,337	No Scopus ID found.
32,180	Still ambiguous (multiple candidates remain).
8,534	Name too common; excluded.
335,768	Total (−2 due to retrieval errors).

B.1.3 Step 3 - Final Candidate Selection and ID Consolidation

Publication-date filter (Step 3a). Scopus profiles are dropped if their publication history is temporally incompatible with the PhD defence. Specifically, we drop a candidate if they have only one publication dated more than five years before the thesis defence, or if their first recorded publication predates the defence by more than ten years. One additional PhD graduate is excluded manually as indistinguishable from a homonym in the data (335,767), leaving 335,767 PhD graduates.

Second-pass field filter (Step 3b). A stricter field-incompatibility check is applied to PhD graduates who still have more than one remaining Scopus candidate. Four sets of cross-field rules flag candidates as errors: (1) hard-science Scopus fields paired with Humanities or Law theses; (2) Social Sciences and Humanities fields paired with STEM theses; (3) the inverse of each; (4) specific cross-discipline combinations (e.g. Economics paired with pure Humanities, Medicine paired with Arts). Flagged candidates are dropped when at least one alternative remains; if only one candidate remains and it is flagged, its Scopus ID is recoded to zero (no match).

After Steps 3a and 3b:

Count	**Description**
146,137	One unique Scopus ID (field- and name-validated).
149,232	No Scopus ID found.
31,864	Still ambiguous (multiple candidates remain).
8,534	Name too common; excluded.
335,767	Total.

Claude-based disambiguation (Step 3c). For the 31,864 PhD graduates with more than one remaining Scopus ID, we use a scoring dataset produced by the Claude AI assistant. For each candidate, Claude was provided with signals including: discipline, country of affiliation, temporal window of publications, first name, and surname. Claude assigned each candidate a matching score and a confidence level (*very_high*, *high*, *moderate*, *low*, or *ambiguous*). Candidates are flagged for exclusion if their score falls more than 0.15 points below the best-scoring Scopus ID for the same PhD. This step eliminates 28,285 spurious Scopus IDs.

After the Claude disambiguation step:

Count	Description
160,923	One unique Scopus ID (field- and name-validated).
149,232	No Scopus ID found.
17,078	Still ambiguous (>1 candidate remaining).
8,534	Name too common; excluded.
335,767	Total.

A *trust* variable is then constructed to indicate the reliability of the resulting match. Trust scores are assigned at this stage and subsequently updated at the publication-merging stage (see paragraph "Final dataset" below).

Trust	Definition	Count
1	Single candidate, or confidence *very_high* or *high* with ≤ 5 IDs	316,791
	of which: no Scopus ID (productivity = 0)	149,232
	of which: at least one validated Scopus ID	167,559
2	Confidence *moderate*, *low*, or *ambiguous* with ≤ 5 IDs	8,795
3	More than 5 remaining candidates, or *ambiguous* with > 2 IDs	10,181
	Total	**335,767**

ID consolidation (Step 3d). It is common for a single researcher to appear under multiple Scopus profiles due to name variants or platform errors. When a PhD graduate has more than one validated Scopus ID after all previous steps, profiles are consolidated into a single identifier.

Final dataset. All 335,767 PhD graduates are retained with one row each. Each observation carries between 1 and 14 Scopus IDs depending on the consolidation outcome, and a trust score of 1, 2, or 3 indicating the reliability of the Scopus match. At the publication-merging stage, an additional 4,576 PhD graduates are reclassified to trust score 3 from score 1. These are graduates whose validated Scopus ID appears in the profiles of more than two other PhD graduates in the sample, indicating a common name not detected by earlier filters and applying the same criterion as for supervisors (Section B.2). We restrict our analysis to PhD graduates with a trust score of 1, representing 312,215 PhD graduates - of whom 162,983 have at least one validated Scopus profile and 149,232 are treated as non-publishers.

B.2 Supervisors

The matching procedure for supervisors follows the same broad logic as for PhD graduates, as detailed in the following subsections.

B.2.1 Step 1 - Retrieval of Author IDs

We submitted a query to the Scopus platform using the name and surname of each supervisor recorded in theses.fr, following the same procedure as for PhD graduates (Section B.1.1). The initial request covered 89,826 unique supervisors. Due to retrieval errors, 2 entries could not be returned, leaving a working sample of 89,824 supervisors. The matching procedure follows the same logic as for PhD graduates, with one important difference: given that excluding a supervisor entails dropping all PhD graduates under their supervision, ambiguous cases are resolved through manual review rather than automated exclusion rules, so as to minimise sample attrition.

After this first retrieval, supervisors are distributed across four categories:

Count	**Description**
52,778	One unique Scopus ID matched.
8,809	No Scopus ID found.
26,322	Between 2 and 23 possible Scopus IDs (ambiguous match).
1,915	Query returned more than 23 results (name too common).
89,824	Total supervisors in working sample.

Field-based selection (Step 1a). The field-filtering procedure follows the same logic as for PhD graduates: Scopus subject codes are recoded into broader disciplinary buckets and compared against the field of the theses supervised by each supervisor in theses.fr.

This step reduces the number of unique Scopus candidate IDs from 149,700 at entry to 96,926. The distribution of supervisors becomes:

Count	Description
55,977	One unique, field-validated Scopus ID.
16,013	No Scopus ID found (including profiles excluded by field mismatch).
15,919	Still ambiguous (multiple field-compatible profiles remain).
1,915	Name too common; excluded from further matching.
89,824	Total.

B.2.2 Step 2 - Retrieval of Publications

We retrieved the full publication history for all supervisors with at least one field-validated Scopus candidate. During retrieval, 174 supervisors could not be returned due to mis-retrieval errors and are dropped, reducing the working sample to 89,650 supervisors.

After retrieval and a second-pass field filter - applied using the same cross-field incompatibility rules as in Step 3b of the PhD pipeline - the distribution becomes:

Count	Description
56,722	One unique Scopus ID (field-validated).
16,206	No Scopus ID found.
14,907	Still ambiguous (multiple candidates remain).
1,915	Name too common; excluded.
89,650	Total.

Manual cleaning. Supervisors with more than six remaining Scopus candidates are reviewed manually. Two categories of supervisors are excluded at this stage: those whose profiles are indistinguishable from homonyms, and those whose validated Scopus ID appears in the profiles of more than two other supervisors in the sample - indicative of a common name not detected by earlier filters and applying the same criterion as for PhD graduates (Section B.1.1). Together, these two exclusions bring the common-name category from 1,915 to 2,904, reducing the working sample to 89,646 supervisors.

Count	Description
70,538	At least One Scopus ID
16,204	No Scopus ID found.
2,904	Name too common; excluded.
89,646	Total.

Final dataset. After this step, the final dataset contains 70,538 supervisors with at least one publication, to which the 16,204 non-publishers are appended, for a total of 86,742 supervisors retained in the dataset.

B.3 Cleaning of publications

The matched Scopus profiles yield 3,152,434 unique publications for supervisors and 2,226,002 unique publications for PhD graduates. Of these, 1,398,180 publications appear in the profiles of both a supervisor and at least one PhD graduate in the sample. After deduplication, the combined pool contains 3,960,499 unique publications.

We exclude publication types that are too infrequent or not analytically relevant: *Report*, *Retracted*, *None*, *Conference Review*, *Business Article*, *Preprint*, and *Abstract Report*. This removes 1,005 publications, leaving a final corpus of 3,959,494 publications distributed across 11 types, as shown in Table B18.

Table B18: Distribution of publications by type

Publication type	Freq.	Percent
Article	2,796,440	70.63
Book	9,692	0.24
Book Chapter	95,215	2.40
Conference Paper	714,214	18.04
Data Paper	1,104	0.03
Editorial	54,055	1.37
Erratum	21,984	0.56
Letter	40,021	1.01
Note	25,189	0.64

Review	177,988	4.50
Short Survey	23,592	0.60
Total	3,959,494	100.00

Co-authorship information. For each publication, we retrieved the full list of co-authors from Scopus. Co-authorship data is missing for 74,064 publications (out of 3,905,187 for which retrieval was attempted). For these publications, the number of co-authors is imputed using the individual-level average number of co-authors computed over all other publications by the same researcher.

B.4 Journal Quality from Clarivate

We match publications to journal-level quality indicators using data from Clarivate (retrieved in 2022). The Clarivate dataset covers 21,762 journals and provides, for each journal: the Article Influence Score (AIS), Eigenfactor, normalised Eigenfactor, number of citable items, cited half-life, citing half-life, and total number of articles. Journal names are standardised in both datasets prior to matching.

Of the 3,979,251 publications in the corpus at this stage, 2,469,322 (62.1%) are successfully matched to a Clarivate journal, while 1,509,929 (37.9%) remain unmatched. The match rate varies substantially by publication type, as shown in Table B19: it is highest for Articles (77.6%) and Reviews (66.7%), and near zero for Books and Book Chapters, which are not indexed in Clarivate.

Table B19: Publications matched to Clarivate journal quality data

Publication type	**Matched**	**Unmatched**	**Total**
Article	2,180,946	631,735	2,812,681
Book	20	9,756	9,776
Book Chapter	412	95,404	95,816
Conference Paper	61,690	652,769	714,459
Data Paper	1,035	69	1,104
Editorial	26,168	28,401	54,569
Erratum	19,940	2,189	22,129
Letter	30,924	9,454	40,378

Note	15,838	9,606	25,444
Review	119,538	59,555	179,093
Short Survey	12,811	10,991	23,802
Total	2,469,322	1,509,929	3,979,251

For publications without a Clarivate match, the Article Influence Score is set to zero. Quality indicators reflect the state of the Clarivate database at the time of download (2022) and are therefore held constant throughout the panel.

B.5 Final Sample

Following the merge of the PhD graduate dataset with the Scopus publication data, and restricting to PhD graduates with a trust score of 1, the sample contains 312,215 PhD graduates. We then apply four successive exclusions to arrive at the final analytical sample. First, 136 PhD graduates are dropped because co-authorship information could not be constructed for any of their publications. Second, 7,712 PhD graduates are dropped because their supervisor has no recorded publications in Scopus; since supervisor productivity enters the analysis as a treatment or control variable, observations for which it cannot be constructed are excluded. Third, 18,001 PhD graduates are dropped because their gender could not be determined from available name databases (Section A.1). Fourth, 11,422 PhD graduates are dropped because their supervisor's gender is missing. These exclusions reduce the sample to a final total of 274,944 PhD graduates who defended their thesis between 1988 and 2021.

C Figures and Tables

Table C1: List of the 21 disciplines

Disciplines	Observations
Humanities and Law	**55,977**
History and Archaeology	10,352
Journalism, Librarianship and Curatorial Studies	1,499
Language and Culture	16,536
Law, Justice and Law Enforcement	16,172
Philosophy and Religion	5,249
The Arts	6,169
Biological and Earth Sciences	**48,340**
Biological Sciences	35,692
Earth Sciences	12,648
Sciences, Technology and Engineering	**121,207**
Agricultural, Veterinary and Environmental Sciences	1,992
Chemical Sciences	17,639
Engineering and technology	39,867
Information, computing and Communication Sciences	18,884
Mathematical Sciences	9,144
Physical Sciences	33,681
Social Sciences	**49,420**
Architecture, Urban Environment and Building	1,261
Behavioural and Cognitive Sciences	12,076
Commerce, Management, Tourism and Services	7,095
Economics	9,282
Education	3,180
Policy and Political Science	3,774
Studies in Human Society	12,752
Entire Sample	**274,944**

C.1 Undivided Publications and Impact

Table C2: Descriptive statistics - PhD Graduates

	All Fields	Humanities & Law	Biological & Earth Sciences	STEM	Social Sciences	Economics
Before 4 years						
Female	0.40	0.49	0.50	0.29	0.48	0.37
Any Publi	0.45	0.15	0.50	0.66	0.25	0.27
Observations	250,889	51,692	44,289	109,867	45,041	8,504
if Any Publi = 1						
Undivided publis	7.53	2.26	8.71	8.40	3.04	3.25
Publications	1.72	1.73	1.40	1.82	1.71	1.75
Impact	0.74	0.17	1.63	0.59	0.36	0.68
Coauthors	9.33	0.89	9.17	11.44	1.59	1.60
Observations	113,067	7,522	22,310	72,174	11,061	2,301
Between 5 and 10 years						
Female	0.39	0.48	0.48	0.28	0.47	0.37
Any Publi	0.34	0.15	0.39	0.46	0.23	0.25
Observations	192,441	40,233	34,718	83,548	33,942	6,625
if Any Publi = 1						
Undivided publis	9.74	2.66	9.74	11.95	4.08	4.77
Publications	2.20	1.97	1.51	2.50	2.07	2.23
Impact	0.80	0.16	1.81	0.63	0.38	0.71
Coauthors	11.63	1.23	11.07	15.32	2.17	1.85
Observations	65,546	5,845	13,639	38,361	7,701	1,667

Table C3: Raw gender gaps

	All Fields	Humanities and Law	Biological and Earth Sc.	STEM	Social Sciences	Economics
			Before t+4			
if Any Publi = 1						
Undivided Publications	**-0.0889*****	**-0.129****	**-0.301*****	**-0.129*****	**-0.215*****	**-0.311*****
	(0.0173)	(0.0546)	(0.0252)	(0.0107)	(0.0322)	(0.0644)
Undivided Impact	**-0.0136**	**0.000154**	**-0.0220***	**0.0371*****	**-0.0292**	**-0.0285**
	(0.00960)	(0.0231)	(0.0130)	(0.0114)	(0.0211)	(0.0400)
# PhD graduates *if Any Publi = 1*	113067	7522	22310	72174	11061	2301
			Between t+5 and t+10			
if Any Publi = 1						
Undivided Publications	**-0.164*****	**-0.174*****	**-0.389*****	**-0.207*****	**-0.240*****	**-0.315*****
	(0.0230)	(0.0486)	(0.0257)	(0.0183)	(0.0397)	(0.0655)
Undivided Impact	**-0.0219****	**-0.0106**	**-0.0164**	**0.0537*****	**-0.00152**	**-0.0169**
	(0.0107)	(0.0238)	(0.0168)	(0.0134)	(0.0319)	(0.0945)
# PhD graduates *if Any Publi = 1*	65546	5845	13639	38361	7701	1667

Notes: The table reports raw gender gaps in publication outcomes for PhD graduates, estimated conditional on having at least one publication (Any Publi = 1), separately for two time windows: before t+4 and between t+5 and t+10, where t denotes the year of PhD graduation. Undivided Publications and Undivided Impact are alternative measures that, unlike the baseline specifications, are not divided by the number of co-authors on each publication. All coefficients report the female dummy coefficient from OLS regressions, capturing the raw gender gap relative to male PhD graduates. Standard errors are clustered by Discipline × University and reported in parentheses. Significance levels are defined as follows: $p < 0.1$ *, $p < 0.05$ **, $p < 0.01$ ***.

Table C4: Adjusted gender gaps

	All Fields	Humanities and Law	Biological and Earth Sc.	STEM	Social Sciences	Economics
			Before t+4			
if Any Publi = 1						
Undivided Publications	**-0.208*****	**-0.280*****	**-0.305*****	**-0.150*****	**-0.279*****	**-0.389*****
	(0.0138)	(0.0584)	(0.0270)	(0.0138)	(0.0340)	(0.0586)
Undivided Impact	**-0.0131**	**0.0190**	**-0.0495*****	**-0.000755**	**-0.0279**	**-0.00258**
	(0.00868)	(0.0294)	(0.0121)	(0.0119)	(0.0270)	(0.0506)
# PhD graduates *if Any Publi = 1*	107726	6236	22033	70199	9258	1955
			Between t+5 and t+10			
if Any Publi = 1						
Undivided Publications	**-0.295*****	**-0.275*****	**-0.405*****	**-0.237*****	**-0.296*****	**-0.366*****
	(0.0187)	(0.0540)	(0.0245)	(0.0238)	(0.0351)	(0.0376)
Undivided Impact	**-0.0102**	**-0.0233**	**-0.0448****	**0.0112**	**-0.0130**	**-0.0504**
	(0.0112)	(0.0220)	(0.0180)	(0.0173)	(0.0247)	(0.0877)
# PhD graduates *if Any Publi = 1*	60947	4789	13400	36594	6164	1373

Notes: The table reports adjusted gender gaps in publication outcomes for PhD graduates, estimated conditional on having at least one publication (Any Publi = 1), separately for two time windows: before t+4 and between t+5 and t+10, where t denotes the year of PhD graduation. Undivided Publications and Undivided Impact are alternative measures that, unlike the baseline specifications, are not divided by the number of co-authors on each publication. All specifications include Discipline × Year × University fixed effects, as well as controls for co-supervision status, supervisor's publication quality, and supervisor gender. The sample is slightly restricted relative to the unconditional table due to the removal of singleton observations arising from the fixed effects structure. Standard errors are clustered by Discipline × University and reported in parentheses. Significance levels are defined as follows: $p < 0.1$ *, $p < 0.05$ **, $p < 0.01$ ***.

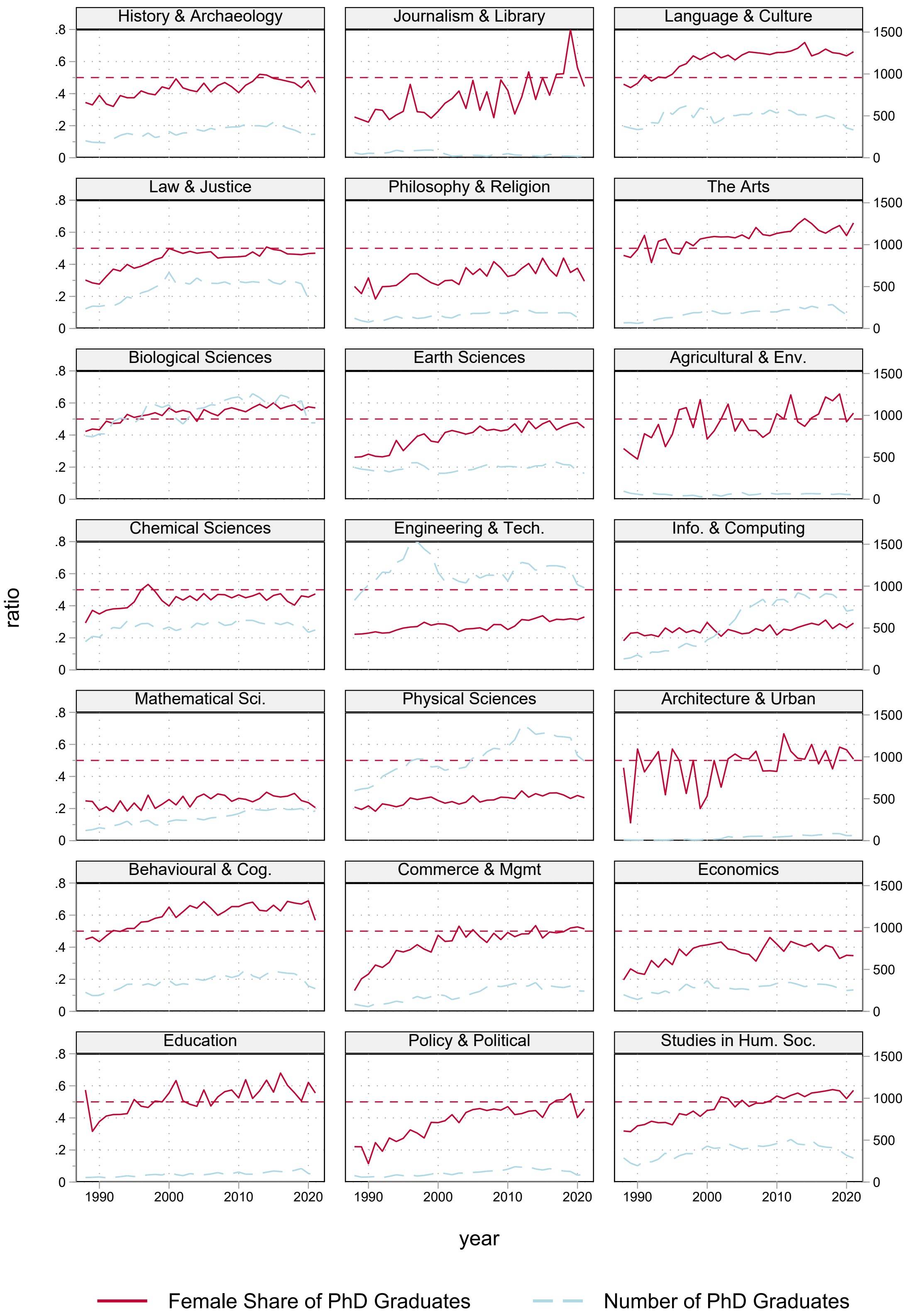

Figure C1: Share of female PhD graduates by fields

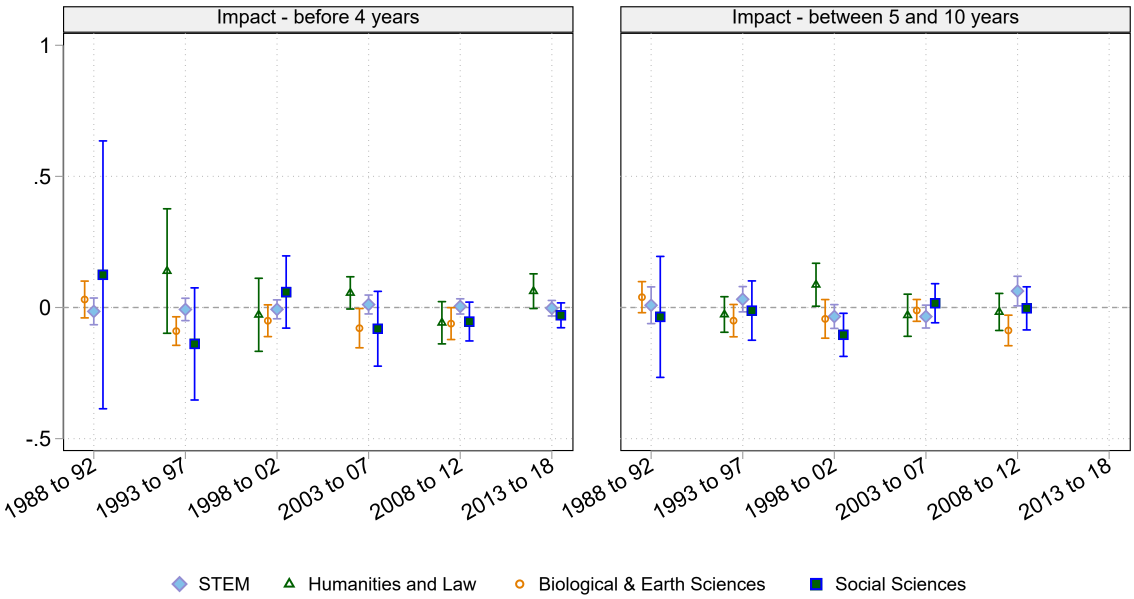

Figure C2: Evolution of gender gaps over time